ARNAV VERMA
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EDUCATION

New York University, Courant Institute of Mathematical Sciences — **New York, NY**
Master of Science - Computer Science — Sept 2026 - May 2028
Courses: Algorithms, Operating Systems, Machine Learning

University of North Texas, College of Engineering — **Denton, TX**
Bachelor of Science - Computer Science, Minor – Mathematics; **GPA 3.70** — Jan 2023 – May 2026
Courses: Algorithms, Linear Algebra, Natural Language Processing, Database Systems, Artificial Intelligence
Awards: President's List, Dean's List, UNT International Affairs Scholarship, UNT Undergraduate Research Fellowship

SKILLS SUMMARY

***Languages**:*	Python, JavaScript, TypeScript, SQL, Bash, Rust
ML/AI:	LangChain, LangGraph, LangSmith, CLIP, FAISS, TensorFlow, Keras, HuggingFace Transformers, RAG, embeddings, Prompt engineering, LLM APIs, AI agents, BERT, FineTuning
Frameworks & Tools:	Flask, React, Docker, Kubernetes, Git, MCP, Github Actions, REST, GraphQL
Data:	PostgreSQL, MySQL, SQLite, ETL, Data Pipelines, Pinecone

PROJECTS

Semantic Video Search Engine | Python, TypeScript, CLIP, FAISS, FastAPI, Docker — ***Blog* | *Code***

- Built a CLIP-based (ViT-B/32) natural-language video search engine with FAISS IndexFlatIP exact retrieval, achieving 11ms p50 / 12ms p95 search latency (~40× under a 500ms target) across 205 videos / 4,415 frames
- Designed scene change aware frame sampling (HSV histogram and Bhattacharyya distance, threshold=0.35) that improved Recall@5 by 14.3 points over fixed-interval sampling validated with a constructed A/B corpus (16 shots, 8 sub-second) after a naive test corpus produced an uninformative tie
- Built crash-safe ingestion with strict write-ordering (index before commit, atomic os.replace) and a write ahead marker for lock free index compaction, guaranteeing recovery from any crash point without vector/metadata desync
- Benchmarked ViT-B/32 vs. ViT-L/14 (+7 Recall@5 points at 2× latency, 512-d to 768-d) and near-duplicate collapsing windows (3-5s optimal) against 225 unit tests and a two-tier Recall@K harness (12 hand labeled, 187 weak labeled queries)

Polish – AI Powered Document Editor | Next.js, TypeScript, Express, PostgreSQL, Redis — ***Code* | *Demo***

- Built an in editor AI assistant on Google Gemini 2.5 Flash that returns structured diffs alongside prose, parsed with regex and markdown fence stripping, with an empty array fallback if the model's output isn't valid JSON, so a bad response shows no suggestions instead of crashing the UI
- Designed a dual token JWT scheme (15-min access / 7-day refresh, distinct signing secrets) backed by a PostgreSQL session table, enabling server-side revocation that stateless JWTs alone can't provide, validated across 5 auth middleware test scenarios
- Built a normalized 5 table PostgreSQL schema with Prisma ORM with autosave debounced at 3s, version snapshots on every meaningful edit, and line level diff comparison across arbitrary version pairs
- Shipped a 3 job GitHub Actions pipeline that typechecks, builds frontend and backend keeping all commits to main safe

EXPERIENCE

Undergraduate Research Fellow – University of North Texas — *May 2024 – May 2026*

- Conducted a systematic multi dimensional generalization study of LLMs (LLaMA, Qwen, Mistral: Dense and MoE) across42 model configurations (3 families × 2 architectures × 7 reasoning strategies) on time and event expression extraction across 4 generalization dimensions: domain shift, adversarial robustness, compositionality, structural length
- Built evaluation pipelines using Hungarian word level LCS matching for F1 scoring, adversarial datasets via SeqAttack(TextFooler/DeepWordBug), and a clinical domain-shift corpus (i2b2 T2C) revealing negative Spearman rank correlation($\rho = -0.15$) between clinical and all other dimensions for events extractions
- Fine tuned RoBERTa, T5, and GPT-2 baselines for time extractions and event extractions sequence labeling (BIOtagging, HuggingFace Trainer); showed fine-tuned PLMs hold more stable cross-dimensional rankings than prompted LLMs despite lower peak performance

PUBLICATIONS

- Iqbal, F., **Verma, A.**, Dutt, R., Das, S., Ray Choudhury, S. "Evaluating Multi-Dimensional Generalization of Large Language Models in Temporal Extraction Tasks." **Under review** at A* conference
- **Verma, A.**, Iqbal, F., Ray Choudhury, S. " Do Large Language Models Reason Causally, or Just Pattern Match? A Multi-Strategy Evaluation" **Under review** at A* conference